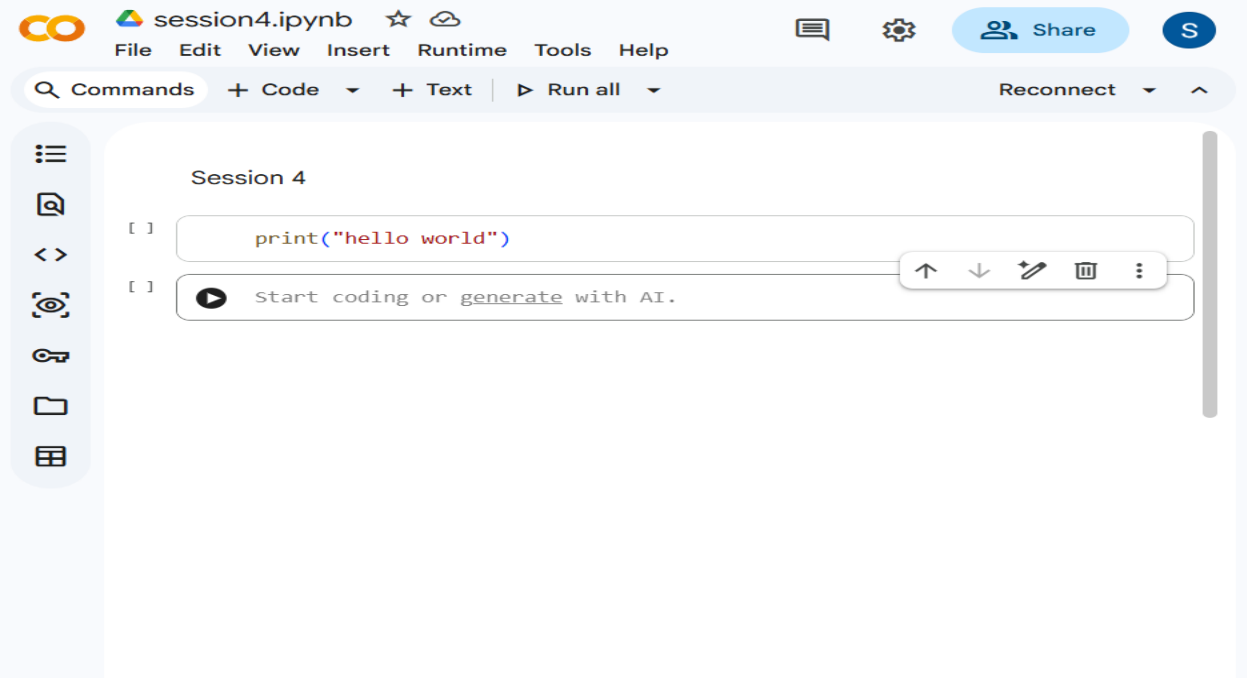
Practical 4

Task 1: Google Colab Setup

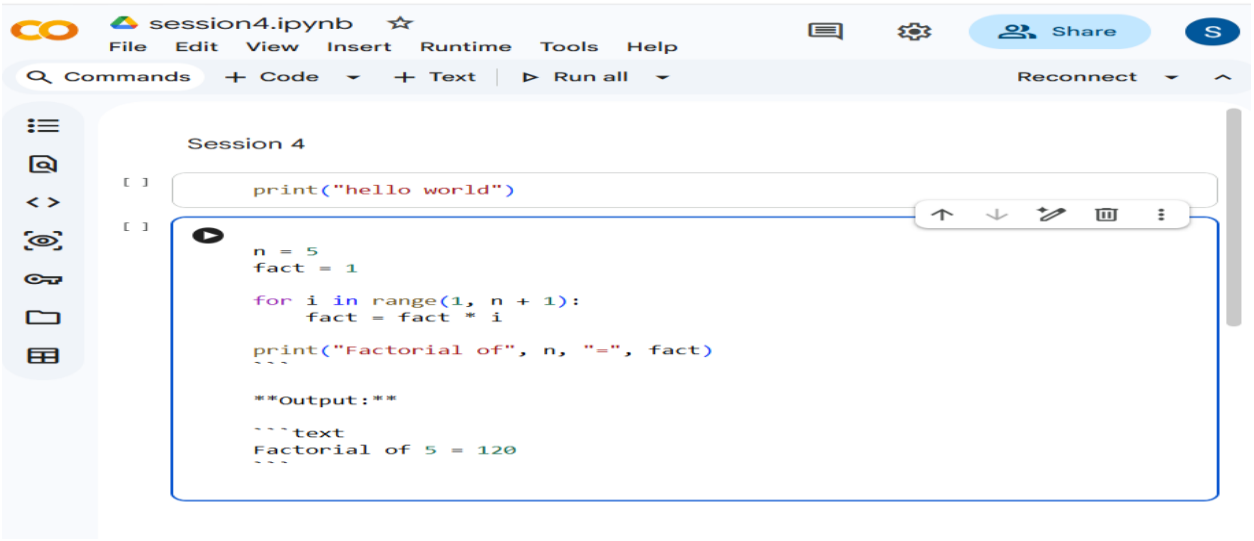


Task 2: C vs Python: Four Structural Differences

No.	Topic	C	Python
1	Variable Declaration	Variables must be declared with a data type, such as `int a = 5;`.	Variables do not require an explicit type declaration, such as `a = 5`.
2	Block Structure	Curly braces `{}` define blocks of code.	Indentation defines blocks of code.
3	Memory Handling	Memory can be managed explicitly using functions such as `malloc()` and `free()`.	Memory management is largely automatic.
4	Grouped Data	A `struct` groups related data fields with declared types.	A dictionary stores data as key-value pairs and allows flexible keys and values.

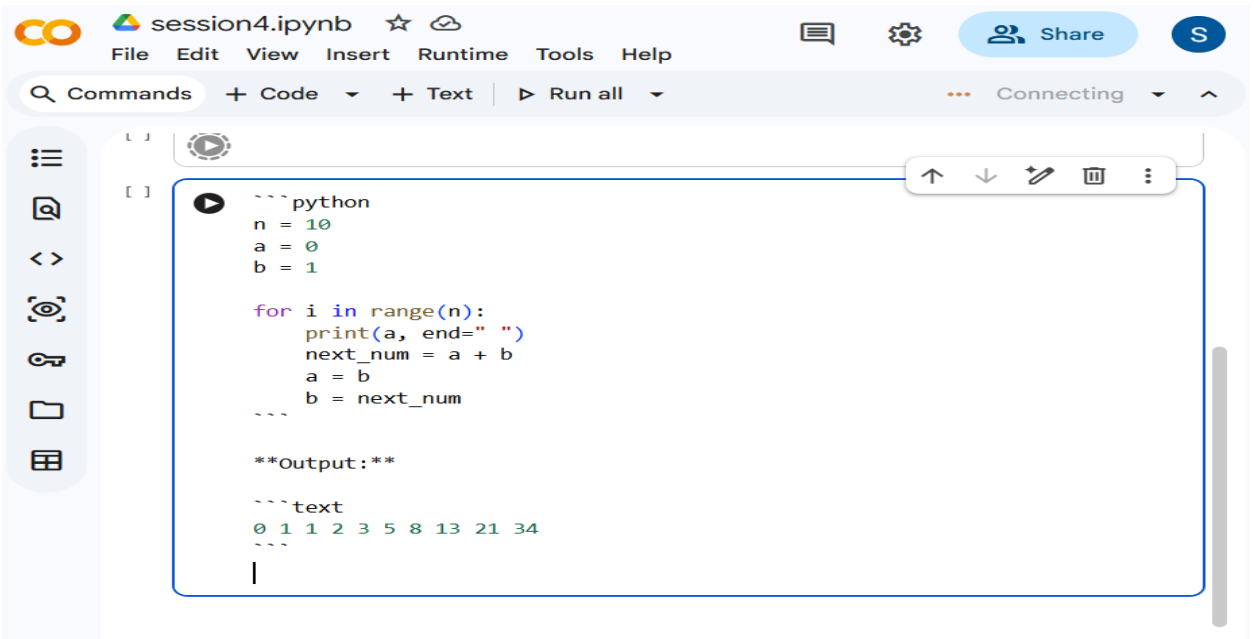
Task 3: Program 1 — Factorial

Write a Python program to calculate the factorial of 5.



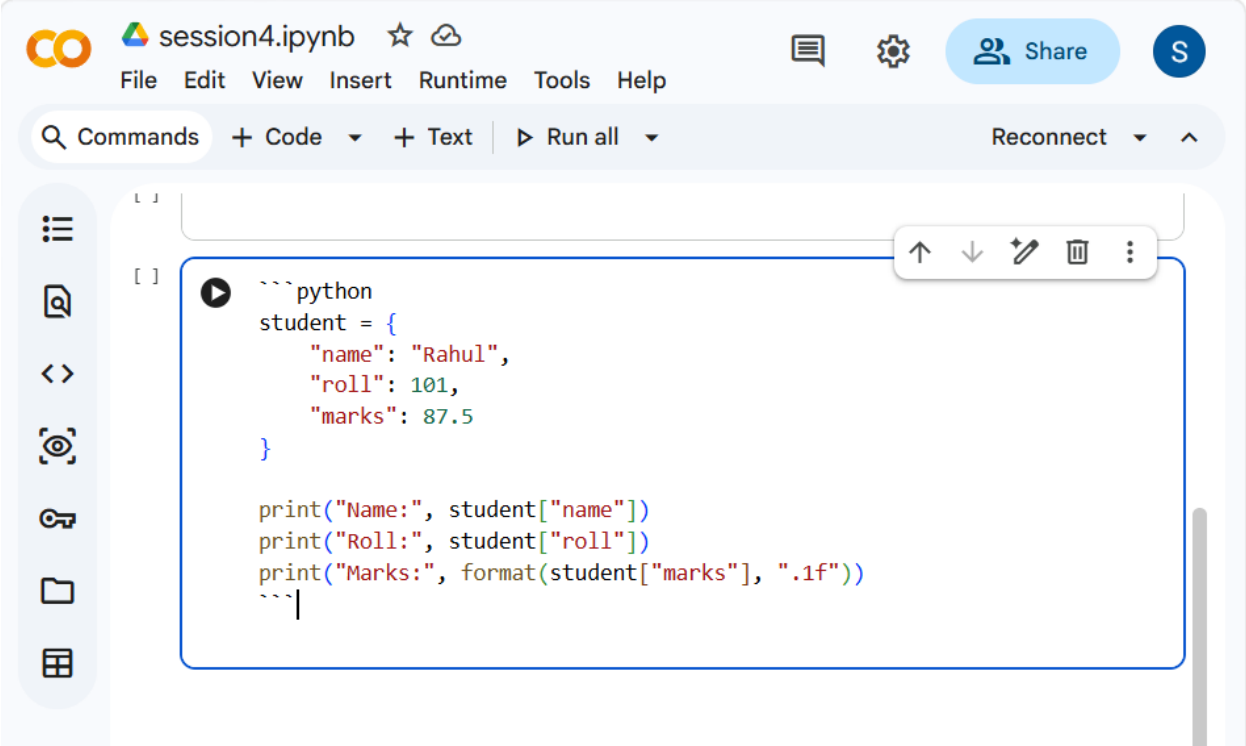
Task 4: Program 2 — Fibonacci Series

Write a Python program to print the first 10 Fibonacci numbers.



Task 5: Program 3 — C Struct to Python Dictionary

Aim: To represent student information using a Python dictionary and display it in the required format.



Task 6: Conceptual Questions

Question 1

What happens when a new key is added at runtime to a Python dictionary, and why is that useful and risky?

When a new key is added to a Python dictionary at runtime, the dictionary stores the new key-value pair without requiring a fixed structure to be declared in advance.

For example, we can add a new key `"grade"` with the value `"A"` to a student dictionary.

This is useful because dictionaries are flexible and allow us to store additional information whenever needed.

However, it can be risky because accidentally using an incorrect key name may create an unwanted entry or cause a `KeyError` when accessing a key that does not exist.

Question 2

Where does the value type come from in a Python dictionary?

In a Python dictionary, the type of each value is determined by the value assigned to it. We do not need to declare the data type of each value explicitly.

For example, a dictionary can store a student's name as a string, roll number as an integer, and marks as a floating-point number.

Python determines the type of each value automatically at runtime.

Task 7: Difference Log

To compare how C and Python handle variables, loops, output and grouped data.

What I Changed	How C Did It	How Python Did It
Variable declaration	Variables require explicit data type declarations, such as `int a = 5;`.	Variables do not require explicit type declarations, such as `a = 5`.
Loop syntax	A `for` loop uses initialization, condition and update, such as `for (int i=0; i<5; i++)`.	A `for` loop commonly iterates over a sequence, such as `for i in range(5):`.
Printing output	The `printf()` function is used to display output.	The `print()` function is used to display output.
Grouping data (struct)	A `struct` groups related fields with declared data types.	A dictionary groups information into key-value pairs.
One thing that surprised me	C requires more explicit syntax for variable declarations and loops.	Python has simpler syntax, and indentation defines code blocks.

Task 8: Peer Verification

S. No.	Program	Expected Output	Verification Status
1	Factorial	Factorial of 5 = 120	Output matches
2	Fibonacci Series	0 1 1 2 3 5 8 13 21 34	All 10 numbers match
3	Student Dictionary	Name: Rahul; Roll: 101; Marks: 87.5	Output matches

Partner's Initials: To be filled by the partner.

Verification Date: 22 October 2026